GHG Statement Report

Version 1.6

Information requirements for verification are as follows:

- GHG Statement Report (this document) containing information relating to the GHG Statement
- GHG Statement calculations (provided separately)
- Supporting information including copies of raw data used

An appropriate level of information should be provided with a clearly referenced and transparent audit trail of decision making, assumptions, explanations, such that a verifier can trace all inputs, outputs and decision making. All data sources and assumptions must be clearly referenced, transparent and traceable.

After this document has been completed for the first time, it may be acceptable to refer back to relevant sections for subsequent Reporting Periods, for example if there have been no changes to the GHG Assessment scope, data collection methodology, calculation methodology or assumptions.

There is no word limit for answers in sections below.

Section A – General Project Information

General Information	
Name of practitioner who prepared the GHG Statement, and relevant competencies. This should include an overview of relevant qualifications and experience undertaking GHG assessments.	Max Lavine Charm Industrial Protocol and Verification Lead
Name of Project / link to Project page on Isometric Registry	https://registry.isometric.com/account/mrv
Date of report	2/20/25
Please specify the Reporting Period this GHG Statement has been prepared for (the Reporting Period describes the time period over which the carbon removal activity assessed in the GHG Statement occurred)	6/1/24-12/6/24 (not including R+D period 7/15/24 - 10/28/24)
Relevant project information if not included in PDD	

Details and links to any supporting documentation	All raw data and supporting documentation has been submitted either 1) directly to Isometric via API, or 2) directly to the Verifier
Please confirm that the GHG Statement has been undertaken in accordance with the information provided in the project boundary section of the PDD. Provide any additional information relating to the GHG Statement project boundary here.	Confirmed

Section B – Project Details

B. GHG Statement - Project Details
Provide information on the Project design and implementation concerning the reporting period. *This should include the following information as a minimum:* ● *Key features of the Project including a technical description* ● *information on the implementation status of the project, For the description of the installed technologies, technical processes and equipment, include diagrams, where appropriate*
6/1-7/15 - Unclaimed Injections During the period from 6/1-7/15, Charm's injection pre-processing experienced technical issues caused by the phase separation of bio-oil in processing. This separation caused the team to be unable to fully empty the blending tank due to offtake pathways being plugged off by extremely viscous oil. This, in turn, confounded our method for quantifying the C content of our injectate which is computed by testing sparged oil, before the addition of other consumables used in processing, and then subtracting the mass of those consumables from the total injectate mass. The relationship between what was introduced to blending and what was removed from blending was disrupted. As a result, the C content of the sparged oil could no longer be reliably used to quantify our removals. After working for several weeks to resolve the issue, our Injection Operations team concluded that (a) we are unable to reliably quantify these injections, and thus only recognize the emissions (v. net removals) and (b) it was necessary to undertake a wholesale revision of our process in order to reliably quantify and execute injections and removals going forward. Charm will not be claiming any credited removals for these injections due to the carbon quantification issues discussed above. However, all emissions generated by these activities are accounted for as site emissions and netted against removals from this period. This ensures a complete accounting for the impacts of Charm's activities during this time.

7/15-10/29 - R&D Period

In consultation with Isometric, we developed and executed a plan to revert the site to “R+D,” pausing its status as an active project. This cut-over took place on 7/15/24. The emissions from all removal activities from 6/1/24-7/14/24 have been accounted for as Site Emissions and have been documented as part of the reporting package delivered to the VVB and Isometric.

This R+D Period officially closed on 10/29/24 following confirmation that the team onsite had developed and rigorously tested a reliable modified process. The emissions from the 6/1-7/15 period have been netted against removals occurring during the reporting period covered by this GHG Statement.

Equipment, Embodied Emissions & Bio-Oil Waste Updates

Additional equipment was added to the site and the amortized embodied emissions of the equipment are included in the Net CDR calculations for this reporting period. This includes 66x 275 gal IBC totes, 63x 330 gal IBC totes, 3x storage tanks and tank bases, and 8x sliding vane pumps, a diesel storage tank, and a pressure washer.

All of the above changes are reflected in additions to the site embodied emissions and are reflected in the Net CDR calculations for the reported removals.

This is also the first reporting period which includes accounting for the disposal of bio-oil as a result of site operations. A contained bio-oil spill at the beginning of June required the removal of bio-oil contaminated turf from the site, which was disposed of at a landfill. In addition, during the 6/1-7/15 period, the team on site had to physically remove overly viscous bio-oil from the tanks for storage in IBC totes. The oil was eventually transferred to a tanker truck for landfill disposal. Emissions from both the induced transport and the disposals themselves are included in the Net CDR calculations for this period. All disposals have been executed by licensed contractors in accordance with EPA regulations.

AR-9 Cavern

These injections are Charm’s first verified removals injected into Vaulted’s AR-9 cavern. This is still part of Vaulted’s facility, utilizing their existing equipment and is part of the same cavern system documented in both Charm’s PDD and their own. However, Charm has conducted diligence and documentation of the cavern’s characteristics in accordance with the requirements of the Biomass or Bio-Oil Storage in Salt Caverns module and submitted these documents to Isometric and the VVB.

Power Sources & Renewables

Charm’s pre-processing site has also been connected to grid power, so this is the first reporting period that includes emissions as a result of grid electricity consumption beyond what occurs at Vaulted’s injection site. This is accounted for using the same KS grid emission factors already in use and is measured through daily monitoring of an on-site energy meter.

Finally, this marks the first period in which Charm is using Procured Renewables in our carbon accounting in the form of Book and Claim Units (BCUs) for Renewable Diesel. Because of our

injection operation's small demand profile and location outside of LCFS jurisdiction, substituting fossil diesel for renewable diesel is not practically feasible. Charm has consulted extensively with Isometric to establish guardrails for the use of BCUs which are demonstrably additional and rigorously accounted for according to Roundtable on Sustainable Biomaterials standards. This has the direct effect of displacing fossil diesel that would otherwise be used with a renewable alternative. Only emissions associated with the use of diesel may be affected by the retirement of BCUs. In Charm's current operations, this is limited to transport process emissions and on-site energy generation.

Section C – Project Changes

C.1 Summary of changes

Please provide details of any changes or deviations from the planned implementation, operation, or monitoring of the project activity that occurred during this reporting period. Project Proponents must determine whether the changes are temporary deviations or permanent changes.
Temporary deviations could include deviations from:

a) The monitoring plan mentioned in the approved PDD.
b) The applied Protocol.
c) The applied modules.
d) Data collection procedure.

If there were any temporary changes or deviations corresponding to this reporting period, please include the details of those changes in this section. If the changes do not fall into the above categories, please note them in Section F of the corresponding PDD instead.

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

1. Charm Industrial will begin using Book-and-Claim Units (BCUs) from Renewable Diesel in our Net CDR calculations, in accordance with the eligibility criteria outlined in the Transportation Emissions Accounting module v1.1. Per Section 4.2.2, Charm has elected to use a bespoke calculation approach which has been reviewed and approved by Isometric. Details on the calculation approach are included in Appendix 4 of the PDD
2. AECN has provided an updated LCA for their production, which has superseded the previous LCA in the GHG accounting for this reporting period. This LCA uses production data from H1 2024 and was completed prior to H2 2024.
3. Charm's Kansas pre-processing site has been connected to grid power. This was discussed in the initial GHG Statement Report as an event expected in the future. Monitoring and use of site electricity consumption in GHG calculations is proceeding as originally outlined
4. Accounting for process waste disposal described above requires an emissions calculation approach that accounts for both the transport of the wastes to the disposal site, and the emissions impact of the disposal itself.

a. Calculation of transport emissions for waste disposal are conducted using the same emission factors and methods as all other transport emissions for the project.
b. There are no emission factors available that specifically address the emissions from the disposal of bio-oil and bio-oil contaminated gravel.
    i. In order to estimate the emissions impact of these disposals, we have engaged Mark Mba-Wright, Associate Professor of Mechanical Engineering at Iowa State University and a professional LCA practitioner
    ii. Prof. Wright identified two emission factors as “lower bound” and “upper bound” reference points to estimate the emissions impact of landfilling bio-oil: landfilling of wood waste and landfilling of food waste. These factors both account for the emissions of biogenic GHGs from landfilled organic material – The “lower bound” emissions from landfilled wood waste are lower than could be reasonably expected from landfilled bio-oil, while the “upper bound” emissions from food waste are higher
        1. Wood contains significant biogenic carbon and, in methanogenic conditions, can produce methane during decomposition. However, because large pieces of wood are relatively intransigent to decomposition, the emissions from landfilled bio-oil may be higher than the emission factor from landfilled wood alone. Therefore, the emission factor for landfilling wood waste is considered an appropriate “lower bound” for estimating the emissions impact of landfilled bio-oil
        2. Food waste also contains significant quantities of biogenic carbon. However, the absence of lignin, high moisture content, and presence of sugars and other compounds consumed by bacteria mean that it decays faster and generates significantly more methane than landfilled wood or bio-oil. Therefore, the emission factor for landfilling food waste is considered an appropriate “upper bound” for estimating the emission impact of landfilled bio-oil
        3. Using an average of these “lower bound” and “upper bound” emission factors is an appropriate and conservative method for estimating the GWP of landfilled bio-oil due to known physical properties of the material
            a. Unlike wood, bio-oil does not contain solid lignin. It may, therefore, decompose more quickly than large pieces of wood waste. An appropriate emission factor for landfilled bio-oil, then, should be higher than the emission factor for landfilled wood waste.
            b. As noted in the PDD, the high-heat, low-oxygen conditions of pyrolysis destroy many of the compounds required to sustain the microbial life that drives anaerobic decay. This is confirmed by a

peer-reviewed chemical characterization of pyrolysis bio-oil from multiple feedstocks (Negahdar et al., 2016) and a specific analysis of pyrolysis oil commissioned by Charm. In addition, a significant proportion of the biogenic GHGs originally present in organic feedstocks are exhausted as tailgas during pyrolysis, which is already accounted for in the bio-oil production LCA. The appropriate emission factor for landfilled bio-oil, then, should be lower than the emission factor for landfilled food waste.

i. As an additional measure to ensure a conservative estimate of these emissions, a food waste emission factor from the Ecoinvent database was selected which is higher, at .603 kg CO2e/kg, than the general emission factor for the landfill disposal of biowaste, which is .547 kg CO2e/kg

c. Taking the average of these “lower bound” and “upper bound” emission factors allows for the calculation of a GWP estimate for waste bio-oil. The emission factor used is .34kg CO2e/kg material, which is the average of wood (.083 kg CO2e/kg material) and food (.603 kg CO2e/kg material) wastes

d. This estimate is conservative due to the selection of an “upper bound” value that is higher than other justifiable possibilities.

c. The emission factor for bio-oil contaminated gravel is a composite emission factor constructed by using the emission factor above for bio-oil and the Ecoinvent emission factor for the “Treatment of Waste Gravel”

i. The composition of the bio-oil contaminated gravel is assumed to be 80% gravel, 20% bio-oil

ii. The emission factor for “Treatment of Waste Gravel” is .0122 kg CO2e/kg material. So, this composite factor was computed as (.8*.0122)+(.2*.34) = .078

d. In sum, these emission factors constitute a rigorous, conservative estimate of the emissions impact of the disposal of little-studied materials.The estimates overall are grounded in reputable, peer-reviewed data from the Ecoinvent and Open LCA databases and the best current understandings of the physical character of bio-oil.

C.2 Impact of changes

For each deviation, provide a description of the nature, extent and duration of the non-conforming reporting period, and:

a) *Propose alternative monitoring arrangements for the non-conforming reporting period; or*
b) *Apply the most conservative values approach*

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

Table: 1 Temporary changes to the project

Period of change	*Nature of change*	*Impact of change*
10/29 onwards	Renewable diesel BCUs may be used to neutralize the emissions impact of diesel usage in applicable modes of transport (truck, US rail) and on-site combustion for energy generation.	Net CDR calculations will only include diesel emissions that are not matched by emissions avoided by the verifiable substitution of renewable diesel for fossil diesel outside of an LCFS jurisdiction
10/29 onwards	Use of updated LCA for AECN bio-oil GHG accounting	The quantity of CO2e is slightly higher than the LCA used previously due to some operational anomalies. Therefore, the gross quantity of CO2e removed per mass of bio-oil sequestered will be slightly lower
10/29 onwards	Charm's KS pre-processing site has been connected to grid power	No material impact – Electricity usage will be accounted for as a site emission in the same manner already used for electricity consumed by Charm's operations at Vaulted and discussed in the initial GHG Statement Report
10/29 Onwards	GHG impacts arising from the disposal of bio-oil and related process wastes (e.g. bio-oil contaminated gravel) has been added to Project Net CDR calculations	This update was made in order to ensure the complete GHG impact of project operations is accounted for in Net CDR calculations. The impact of the change, therefore, is that the project maintains complete and comprehensive GHG accounting as operating circumstances evolve.

Section D – GHG Statement Methodology

D.1 GHG Statement Methodology - Data

Provide information on the GHG Statement approach and methodology in relation to data collection.

This should include the following information as a minimum:

- *Data collection procedures - how data was requested and gathered, who supplied each data point, the format data was received in, and the date data points were received*
- *How data quality was assessed and how the data hierarchy was followed in terms of using measured, calculated and estimated data. Evidence and justification should be provided in any cases where measured data was not used*
- *Details of data validation processes followed, including treatment of missing data*

Evidence of all raw data that informed the assessment (including data that informed any assumptions) must be appended to the PDD, or linked within the GHG Statement.

This section only includes content that is new or modified. Please reference Appendix 3 of the PDD for information on the items above concerning other areas of the project

1. Disposal of Process Waste
    a. Fuel Emissions from Waste Transport to Disposal Facility
        i. Data Sources
            1. Cargo mass and destination are taken from manifests provided by EPA-regulated waste disposal providers
            2. The vehicle routing is assessed using Google Maps, using the longest given route in order to ensure a conservative estimate of removals
            3. Vehicle Type is always a Heavy Duty Vehicle due to the quantity and nature of cargo – A flatbed truck is used for solid waste, and a tanker truck is used for liquid waste
            4. This information allows for the calculation of WTW emissions from fuel use using GLEC emission factors
        ii. Data Quality Assessment
            1. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a reputable and reliable source of input data
            2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
            3. GLEC qualifies as a “reputable source” for emission factors given the Isometric Protocol’s definition
    b. Vehicle Embodied Emissions from Waste Transport to Disposal Facility
        i. Data Sources
            1. For all truck transport, the Vehicle Embodied Emissions are calculated using the GREET 2023 emissions factors for Medium-and-Heavy-Duty (MHDV) Trucks and Trailers. The total value for the embodied emissions is divided by the expected useful life in miles for a Heavy Duty compression ignition

(diesel) engine established by US EPA Office of Transportation and Air Quality in 2016 for engines manufactured after 2004. This yields a quantity of embodied carbon per mile traveled.

a. The trucks being used in this project are not used by Charm, but by a third-party transport provider contracted by Charm. Therefore, a method in which only the mileage driven while under contract is considered is necessary.
b. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute (ATRI) analysis

ii. Data Quality Assessment

1. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition
2. The variable of mileage has a low uncertainty, especially given that the routing between points is generally consistent over time and Google Maps route selection controls for uncertainty by selecting the longest route when multiple are presented to ensure a conservative estimate of removals

c. Emissions from Material Disposal and Decomposition

i. Data Sources

1. The quantity of material disposed is taken from manifests provided by EPA-regulated waste disposal providers
2. The emission factor(s) quantifying the emissions impact of the material disposed are sourced from reputable sources including but not limited to the GREET database, the EcoInvent database, Open LCA, USLCI database, ICE database, and/or the Athena Impact Estimator

a. Emission factors are used as-is where feasible. To account for the disposal of less-studied material such as bio-oil, a reasonable estimate may be made using proxy values. In this case, references to baseline values and sources, as well as any additional assumptions and justifications will be documented by the project and are subject to review and approval by Isometric

ii. Data Quality Assessment

1. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a highly reputable and reliable source of input data
2. The LCA database sources listed above qualify as "reputable sources" given the Isometric Protocol's definition

D.2 GHG Statement Methodology - Calculations

Provide information on the GHG Statement approach and methodology in relation to calculations.

This should include the following information as a minimum:

- *General description of the methodology, criteria and procedures used as a basis for*

the assessment, including reference to any documentation (including Isometric protocols and modules), guidance, industry standards and best practice that were followed

- *Information on calculation procedures followed*
- *Information on any tools used as part of the assessment*
- *Procedure for selecting emission factors including how age (age of data, and the period over which they have been collected), geography (the region or country from where the data have originated), technology (whether the data are specific to a particular technology or mix of many), methodology (the approach applied to gather or calculate the data) and competency (proficiency of entity that developed the data) were considered.*
- *Assumptions and limitations - provide full transparency in terms of value-choices, rationales and expert judgements*
- *Details of a sensitivity analysis, how parameters were assessed and evidence behind choices*
- *Details of uncertainty analysis*
- *Details of any uncertainty adjustments (e.g. %) applied in instances of high uncertainty*

This section only includes content that is new or modified. Please reference Appendix 3 of the PDD for information on the items above concerning other areas of the project

1. Disposal of Process Waste
    a. Fuel Emissions from Waste Transport to Disposal Facility
        i. Calculation Methodology
            1. Cargo weight and destination are documented according to manifest from EPA-regulated waste disposal provider and point-to-point distance between origination and disposal facilities from Google Maps are used to calculate the WTW emissions for the journey using the appropriate vehicle type by applying the appropriate GLEC emissions factor (flatbed or tanker truck) to the tons-miles value
        ii. Justification of Variables and Emission Factors Used
            1. Vehicle Type is always a Heavy Duty Vehicle due to the quantity and nature of cargo – A flatbed truck is used for solid waste, and a tanker truck is used for liquid waste
            2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Analysis section
            3. Disposal manifests and providers are subject to regulation and review by USEPA, qualifying them as a reputable and reliable source of input data
            4. GLEC qualifies as a “reputable source” for emissions factors given the Isometric Protocol’s definition
        iii. Unit/Frequency of Calculation
            1. Transport emissions are calculated for each manifested waste disposal required by project activities
        iv. Modifications to Address Uncertainty
            1. Masses are transmitted to Isometric DMRV with the maximum error of a legal-for-trade truck scale for variance propagation, as is the case for other project transport emissions as discussed in the Uncertainty Assessment section of the PDD
            2. When Google Maps routing provides multiple options, the longest routing option is chosen as discussed in the Uncertainty Assessment section of the PDD

b. Vehicle Embodied Emissions from Waste Transport to Disposal Facility
    i. Calculation Methodology
        1. Per-mile embodied emissions are calculated based on GREET emissions levels for the manufacture of medium-and-heavy-duty trucks and trailers and divided by EPA expected useful life in mileage for heavy-duty diesel engines. This calculation yields a per-mile embodied emissions level that can be applied to Charm-specific journeys using vehicles owned by a transport contractor.
        2. The emissions factor is modified to include the average "deadhead" or unloaded journey distance for a vehicle of this type according to American Transportation Research Institute analysis
        3. Per-mile embodied emissions are applied to the waste being transported according to journey distance.
    ii. Justification of Variables and Emission Factors Used
        1. GREET, ATRI, and the US EPA qualify as "reputable sources" given the Isometric Protocol's definition of "A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body)."
        2. Google Maps routing is a reliable source of mileage information and is controlled for uncertainty as described in the Uncertainty Assessment section
        3. Calculation of embodied emissions by mile is necessary due to the fact that the trucks may be used by the transport provider for different routes and applications over time. Therefore, a calculation method is appropriate that only takes into account the mileage associated with Charm journeys.
    iii. Unit/Frequency of Calculation
        1. Transport emissions are calculated for each manifested waste disposal required by project activities
    iv. Modifications to Address Uncertainty
        1. When Google Maps routing provides multiple options, the longest routing option is chosen as discussed in the Uncertainty Assessment section of the PDD
c. Emissions from Material Disposal and Decomposition
    i. Calculation Methodology
        1. Total mass of material disposed is scaled by appropriate emission factor as indicated by the disposal method and material
    ii. Justification of Variables and Emission Factors Used
        1. The mass, material, and the disposal method are documented in the manifest and disposal documents provided by the EPA-regulated waste disposal provider, a reputable and reliable source for these datapoints
        2. The emission impact of disposal is quantified using emissions factors from reputable sources including but not limited to the GREET database, the EcoInvent database, NAICS supply Chain Emission Factors, USLCI database, ICE database, and/or the Athena Impact Estimator, reputable sources based on the Isometric Protocol's definition
            a. Emission factors are used as-is where feasible. To account for the disposal of less-studied material such as

bio-oil, a reasonable estimate may be made using proxy values. In this case, references to baseline values and sources, as well as any additional assumptions and justifications will be documented by the project and are subject to review and approval by Isometric

iii. Unit/Frequency of Calculation
    1. Disposal emissions are calculated for each manifested waste disposal required by project activities at the time that the disposal activity is confirmed by relevant documentation

iv. Modifications to Address Uncertainty
    1. Masses are transmitted to Isometric DMRV with the maximum error of a legal-for-trade truck scale for variance propagation, as is the case for other project transport emissions as discussed in the Uncertainty Assessment section of the PDD

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Emissions from conventional diesel fuel–used in truck transport, rail transport, and on-site energy generation–May be neutralized through the retirement of RD BCUs provided sufficient evidence for economic and environmental additionality, chain of custody, and LCA/carbon accounting. Full details are provided in Appendix 4 of the most recent version of the PDD. The approved calculation method for their use is as follows:

Equation for Well-to-Tank Emissions for Renewable Diesel

$EP_{RD, WTT} = CI_{RD, WTT} * LHV_{RD, WTT}$

Where:

1. $EP_{RD,WTT}$ is the WTT emission profile for each kilogram of a given batch or delivery of RD
2. $CI_{RD, WTT}$ is the WTT carbon intensity given in the Proof of Sustainability (PoS) denominated in grams of CO2e per MJ of fuel
3. $LHV_{RD, WTT}$ is the lower heating value given in the PoS denominated in MJ per kg of fuel

Equation for Well-to-Wheel Emissions for Renewable Diesel

$EP_{RD, WTW} = (EP_{RD, WTT} + EF_{RD, TTW})$

Where:

1. $EP_{RD, WTW}$ is the WTW emission profile for each kilogram of a given batch or delivery of RD
2. $EP_{RD,WTT}$ is the WTT emission profile for each kilogram of a given batch or delivery of RD
3. $EF_{RD, TTW}$ is the TTW emission factor for the applicable type of fuel (currently sourced from GLEC)

Equation for Renewable Diesel WTW Avoided Emissions

$AE_{RD} = EF_{FD, WTW} - EP_{RD, WTW}$

Where:

1. AE_{RD} is the avoided emissions from the use of a given quantity of RD
2. $EF_{FD, WTW}$ is the WTW emission factor quantifying the emission impact in CO2e for each kilogram of fossil diesel (currently sourced from GLEC)
3. $EP_{RD, WTW}$ is the WTW emission profile quantifying the emission impact in CO2e for each kilogram of renewable diesel associated with a given batch or delivery

Equation for Offsetting Fossil Diesel Emissions with Avoided Emissions from Renewable Diesel BCUs

Net $CO2e_{FD}$ = ($CO2e_{Transport}$ + $CO2e_{On\text{-}Site}$) - (AE_{RD} * $Mass_{RD}$)

Where:

1. Net $CO2e_{FD}$ is the net CO2e attributable to the use of fossil diesel which will be deducted from the gross sequestration in a given GHG Statement
2. $CO2e_{Transport}$ is the total CO2e attributable to transport activities, typically calculated using the Distance-Based Method provided in the Transportation Emissions Accounting module
3. $CO2e_{On\text{-}Site}$ is the total CO2e attributable to on-site (non-transport) activities, typically calculated by applying a WTW emission factor to the quantity of fuel delivered to the site for project activities
4. AE_{RD} is the avoided emissions from the use of a given quantity of RD
5. $Mass_{RD}$ is the mass of RD whose attributes have been procured and are being retired by the project in order to offset FD emissions in a given GHG Statement

Note: Net $CO2e_{FD}$ may only be a positive integer or zero. It may never be a negative value.

Section E – GHG Statement Results

GHG Statement Results		
The following information should be provided in this section: ● *Baseline results reported in t CO_2e for the Reporting Period* ● *Net CO_2e removals results reported in t CO_2e for the Reporting Period. These must be aggregated and also broken down into GHG SSRs* ● *If it has not already been covered in the* 'Uncertainty assessment' section of the PDD, provide *outcomes of sensitivity and uncertainty analyses including a statement of how uncertainty affects the results and how it has been addressed to minimize misrepresentation* ● *A statement of how the assumptions and choices made in the assessment are conservative* *The GHG Statement calculations must be provided separately, including all raw data and evidence. The GHG Statement calculations must be clearly referenced with a transparent audit trail of evidence, decision making, assumptions, explanations.*		
	Kg	MT
Gross Bio Oil Sequestered	254310	254.31
	Kg CO2e	MT CO2e

Gross CO2e Sequestered	388510	388.51
Project Emissions	Kg CO2e	MT CO2e
Process Emissions	129600	129.60
Site+Amortized Embodied Emissions	216050	216.05
Uncertainty Discounts	1050	1.05
Emissions Reductions	204910	204.91
Total Emissions for Reporting Period (kg)	346700	
Total Emissions for Reporting Period (MT)	346.70	
Net Removals for Reporting Period (kg)	246730	
Net Removals for Reporting Period (MT)	246.73	

Please reference the PDD for documentation of uncertainty and sensitivity analyses, and an account of system boundaries, monitored data, and calculation methodology. All raw data and evidence regarding these removals has been transferred to Isometric from Charm's Ledger via API. Supplementary data, documentation, and guidance has been provided to the VVB to help guide verification

Section F – GHG Statement Version Changes

GHG Statement Version Changes

For each statement resubmission, please specify the new version date (which is the same as the resubmission date) and the updated total of CO_2e tonnes removed.

GHG Statement Versions

Version date	*Total CO_2e tonnes removed*
2/20/25	246.73
2/7/25	246.73
1/15/25	234.28

Document History

Version	Date	Description
v1.6	June 2024	Revision to include Section F - GHG Statement Version changes, in line with the versioning product experience
v1.5	May 2024	Revision to include section on project changes, and other editorial changes
v1.4	November 2023	Public version